

MA.912.AR.2.2

Benchmark

MA.912.AR.2.2 Write a linear two-variable equation to represent relationships between quantities from a graph, a written description or a table of values within a mathematical or real-world context.

Benchmark Clarifications:

Clarification 1: Instruction includes the use of standard form, slope-intercept form and point-slope form, and the conversion between these forms.

Connecting Benchmarks/Horizontal Alignment

- MA.912.AR.4.1
- MA.912.AR.9.1
- MA.912.F.1.4
- MA.912.FL.3.2, MA.912.FL.3.4
- MA.912.DP.2.4

Terms from the K-12 Glossary

- Linear Equation
- Slope
- y -intercept

Vertical Alignment

Previous Benchmarks

- MA.8.AR.3.2
- MA.8.AR.3.3

Next Benchmarks

- MA.912.NSO.4.2
- MA.912.AR.9.8
- MA.912.AR.9.9

Purpose and Instructional Strategies

In grade 8, students wrote linear two-variable equations in slope-intercept form from tables, graphs and written descriptions. In Algebra I, students write linear two-variable equations in all forms from real-world and mathematical contexts. In future courses, students will write systems and solve problems involving systems in three-variables and linear programming. Additionally, linear equations and linear functions are fundamental parts of all future high school courses.

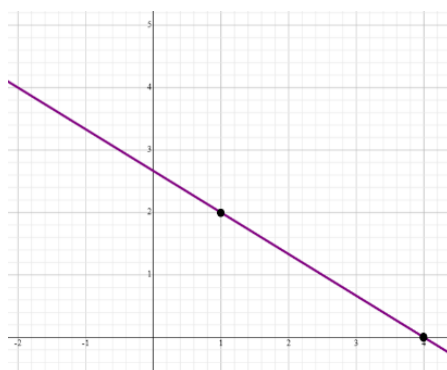
- Instruction includes making connections to various forms of linear equations to show their equivalency. Students should understand and interpret when one form might be more useful than other depending on the context.
 - Standard Form
Can be described by the equation $Ax + By = C$, where A , B and C are any rational number. This form can be useful when identifying the x - and y -intercepts.
 - Slope-Intercept Form
Can be described by the equation $y = mx + b$, where m is the slope and b is the y -intercept. This form can be useful when identifying the slope and y -intercept.
 - Point-Slope Form
Can be described by the equation $y - y_1 = m(x - x_1)$, where (x_1, y_1) are a point on the line and m is the slope of the line. This form can be useful when a point on the line is given and the y -intercept is not easily determinable.
- Look for opportunities to point out the connection between linear contexts and constant rates of change.
- Problem types should include cases for vertical and horizontal lines.

Common Misconceptions or Errors

- Students may have difficulty identifying both variables from a context. Much of their work previously has involved univariate contexts. Place emphasis on asking students what is changing in each context. Help guide their thoughts to recognize bivariate contexts as having two “things” that change in tandem.
- Students may attempt to estimate intercepts in order to continue using a linear form they prefer for some contexts. Use these opportunities to address the need for precision in mathematics.

Strategies to Support Tiered Instruction

- Instruction includes strategies from MA.912.AR.1.2 on rearranging equations to help students when they convert from one form of two-variable linear equation to another.
- Teacher models opportunities to address the need for precision in mathematics when determining intercepts.
 - For example, a student could prefer to use slope-intercept form when writing two-variable linear equations. If the given information, as shown below, is two points that do not include the y -intercept, then the student may only estimate the y -intercept rather than determining it exactly. The student should realize that they could use point-slope form to write the equation without having to determine the y -intercept.



- Instruction includes explicit questions such as “What is staying the same or constant?”, “What is changing or varying?” or “Is there anything else varying?”
 - For example, students are given the situation where a dog groomer charges \$25 for a shampoo and hair cut plus \$10 for each hour the dog stays at the groomer and are asked to write a linear two-variable equation that represents the total cost. The teacher can provide questions to help determine the constant value and two variables.

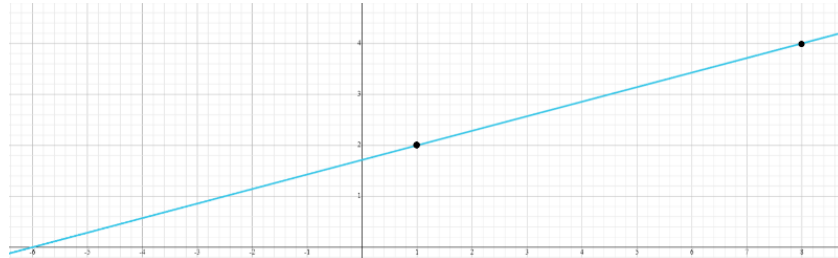
Constant	Variable #1	Variable #2
Charge for Shampoo (\$25)	Number of Hours (\$10 each)	Total Cost

- Instruction includes discussions about what a particular coordinate point on a line means in the context of the problem. Teachers may ask, “What does the identified point represent in the context of the problem?” and “How does the y change as x increases/decreases?”
- Teacher models finding the slope by color coding the points.
 - For example, to find the slope of a line passing through points (1 2 and (4 0 , you would use $m = \frac{2-0}{1-4}$ or $m = \frac{0-2}{4-1}$.
- For students who need extra support in adding or subtracting integers, instruction includes using two-colored counters or algebra tiles to model the operation.
 - For example, given the expression $8 - (-2)$, students can use two-colored counters to find the difference as shown.

Start with 8.	There are not two negatives to subtract so you must add two zero pairs.	Subtract the two negatives to show that $8 - (-2) = 10$.

- Instruction includes a graphic organizer to identify the key features. Based on the key features identified, ask students which form of an equation would be the best.

- For example, given the graph below, students can use an organizer to fill in some of the information.



Once the information is filled in, students can write an equation of the line and determine the rest of the key features.

Slope	x -intercept	y -intercept	Point(s) on the graph that are not intercepts	Equation of Line

Instructional Tasks

Instructional Task 1 (MTR.2.1, MTR.3.1, MTR.5.1)

Jamie bought a car in 2005 for \$28,500. By 2008, the car was worth \$23,700.

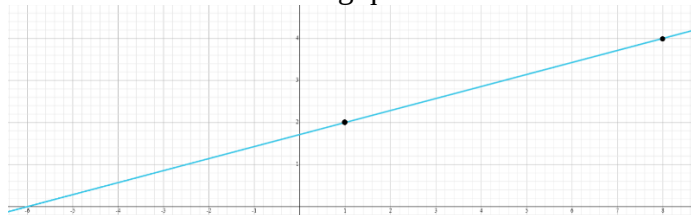
Part A. What function type could model the given situation?

Part B. What is the rate of change in the vehicle's worth per year?

Part C. Create a model that describes this situation.

Instructional Task 2 (MTR.3.1, MTR.4.1)

Use the graph below to answer the following questions.



Part A. In order to write the equation that represents this line, what information do you need?

Part B. Write a linear two-variable equation that represent the graph below. Justify why you choose the linear form to represent the graph.

Part C. Write a real-world situation that could represent this graph.

Instructional Items

Instructional Item 1

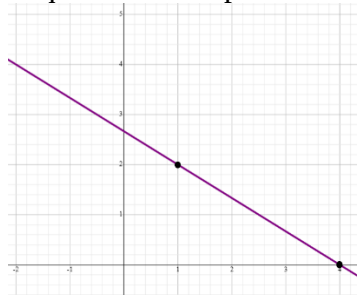
Sharon is ordering tickets for an upcoming basketball game for herself and her friends. The ticket website shows the following table of ticket options.

Number of tickets	Cost (including processing fee)
1	\$42.50
2	\$68.50
3	\$94.50
4	\$120.50
5	\$146.50
10	\$276.50
15	\$406.50

Write a linear two-variable equation to represent the total cost C of t tickets.

Instructional Item 2

Write a linear two-variable equation that represents the graph below.



**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*